

EXNO 1: Estimation of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test Case 1:

Investigating the significance of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line using the following parameters

- a) Resistance = $10 \Omega/\text{Km}$
- b) Capacitance = $8 \mu\text{F}/\text{Km}$
- c) Inductance = $3.5 \text{ mH}/\text{Km}$
- d) Conductance = $4 \mu\text{mho}/\text{Km}$
- e) Frequency = 2 KHz

AIM: Investigating the significance of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line with Resistance = $10 \Omega/\text{Km}$, Capacitance = $8 \mu\text{F}/\text{Km}$, Inductance = $3.5 \text{ mH}/\text{Km}$, Conductance = $4 \mu\text{mho}/\text{Km}$, Frequency = 2 KHz

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clc;
```

```
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
```

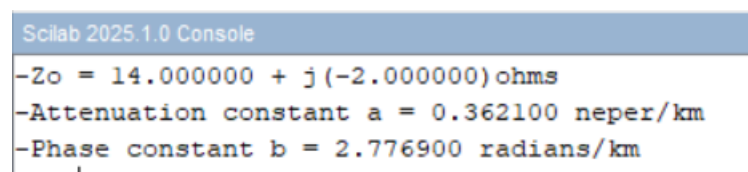
```
w=2*%pi*f;
```

```

Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);

```

OUTPUT:



```

Scilab 2025.1.0 Console
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km

```

RESULT:

The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

Test Case 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line at a frequency $f=2.5$ KHz. Vary the resistance value from $10\ \Omega$ to $50\ \Omega$ at the interval of $10\ \Omega$ with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

AIM: To examine the impact of resistance and inductance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line at a frequency $f=2.5$ KHz. Vary the resistance value from $10\ \Omega$ to $50\ \Omega$ at the interval of $10\ \Omega$ with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant. Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clc;  
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);  
w=2*%pi*f;  
Z=R+(%i*w*L);  
Y=G+(%i*w*C);
```

```

Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);

```

OUTPUT:

Variable R

Constants L=3mH; f=2000; G=.4*(10⁻⁶); C=8*(10⁻⁶).

S.No.	Inputs(R)	Z0(ohms)	α (neper/km)	β (radians/km)
1	10	14.000000 + j(-2.000000)	0.362100	2.776900
2	30	15.000000 + j(-5.000000)	1.026500	2.938300
3	50	16.000000 + j(-8.000000)	1.582800	3.175700
4	70	17.000000 + j(-10.000000)	2.050100	3.432600
5	90	18.000000 + j(-12.000000)	2.453500	3.687700

Variable L

Constants R=10 Ω ; f=2000; G=.4*(10⁻⁶); C=8*(10⁻⁶).

S.No.	Inputs(L)	Z0(ohms)	α (neper/km)	β (radians/km)
1	3*10 ⁻³	14.000000 + j(-2.000000)	0.362100	2.776900
2	4*10 ⁻³	16.000000 + j(-2.000000)	0.314700	3.194600
3	5*10 ⁻³	18.000000 + j(-1.000000)	0.282000	3.565500
4	6*10 ⁻³	19.000000 + j(-1.000000)	0.257700	3.902100
5	7*10 ⁻³	21.000000 + j(-1.000000)	0.238700	4.212300

RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line is examined at a frequency f=2.5 KHz.

Test Case 3:

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8 μ F to 16 μ F at the interval of 2 μ F with R,G and L values constant.

AIM: To explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
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- End the program

PROGRAM:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));
```

```

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);

b=imag(P);

b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);

```

OUTPUT:

Variable f

Constants L=3mH; R=10; G=.4*(10⁻⁶); C=16*(10⁻⁶).

S.No.	Inputs(f) Hz	Z0(ohms)	α (neper/km)	β (radians/km)
1	1000	14.000000 + j(-4.000000)	0.353700	1.421300
2	2000	14.000000 + j(-2.000000)	0.362100	2.776900
3	3000	14.000000 + j(-1.000000)	0.363800	4.145700
4	4000	14.000000 + j(-1.000000)	0.364400	5.518300
5	5000	14.000000 + j(-1.000000)	0.364700	6.892500

Variable C

Constants R=10; L=.009; f=2500; G=.4*(10⁻⁶);

S.No.	Inputs(C) F	Z0(ohms)	α (neper/km)	β (radians/km)
1	16*10 ⁻⁶	24.000000 + j(-1.000000)	0.210700	5.964500
2	17*10 ⁻⁶	23.000000 + j(-1.000000)	0.217200	6.148000
3	18*10 ⁻⁶	22.000000 + j(-1.000000)	0.223500	6.326300
4	19*10 ⁻⁶	22.000000 + j(-1.000000)	0.229600	6.499600
5	20*10 ⁻⁶	21.000000 + j(-1.000000)	0.235600	6.668500

RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line is investigated for various values of frequencies and capacitance values.